

Predictive Analytics Using Historical Data

Project Overview

This project demonstrates the use of Predictive Analytics and Machine Learning techniques to forecast future sales using historical business data. The objective is to analyze historical sales trends, build a predictive model, evaluate its performance, and generate future sales forecasts that can support business decision-making.

The project was developed using Python and Scikit-Learn in Google Colab.

Author

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Domain: Data Analytics & Machine Learning

Problem Statement

Organizations generate large amounts of historical sales data. Analyzing this data manually is difficult and time-consuming. Predictive Analytics helps identify patterns and forecast future outcomes, enabling businesses to make data-driven decisions.

This project aims to predict future sales based on historical sales records using a Linear Regression model.

Objectives

- Analyze historical sales data.
 - Perform data preprocessing and feature engineering.
 - Build a machine learning model for sales prediction.
 - Evaluate model accuracy using standard metrics.
 - Forecast future sales trends.
 - Visualize historical and predicted results.
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Dataset Information

Predictive Analytics Using Historical Data

Features

Feature	Description
Date	Monthly sales date
Sales	Total sales amount
Profit	Profit earned
Quantity	Number of products sold
Discount	Discount percentage

Dataset Summary

- Total Records: 36
- Time Period: 2023–2025
- Missing Values: 0
- Dataset Type: Time-Series Sales Data

Technologies Used

Programming Language

- Python

Platform

- Google Colab

Libraries

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-Learn

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Project Workflow

Historical Dataset



Data Collection



Data Cleaning



Feature Engineering

(Year, Month)



Train-Test Split



Linear Regression Model



Prediction



Model Evaluation

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(MAE, RMSE, R^2)

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Future Forecasting

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Visualization & Results

Machine Learning Model

Algorithm Used

Linear Regression

Input Features

- Profit
- Quantity
- Discount
- Year
- Month

Target Variable

- Sales

Model Evaluation Metrics

The model was evaluated using:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)

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- R^2 Score
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Visualizations

The following visualizations were generated:

- Actual vs Predicted Sales Scatter Plot
 - Actual vs Predicted Sales Line Plot
 - Residual Plot
 - Historical Sales Trend
 - Feature Importance Chart
 - Future Sales Forecast Graph
 - Historical vs Forecasted Sales Trend
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Project Structure

Predictive_Analytics_Project/

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|— data/

| |— sales_data.csv

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|— outputs/

| |— sales_forecast_results.csv

| |— actual_vs_predicted.csv

| |— feature_importance.csv

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|— screenshots/

Predictive Analytics Using Historical Data

- | └─ 01_Project_Setup.png
- | └─ 02_Dataset_Loaded.png
- | └─ 03_Data_Preprocessing.png
- | └─ 04_Model_Training.png
- | └─ 05_Model_Evaluation.png
- | └─ 06_Actual_vs_Predicted.png
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- | └─ 08_Residual_Plot.png
- | └─ 09_Future_Sales_Forecast.png
- | └─ 10_Feature_Importance.png
- | └─ 11_Historical_Sales_Trend.png
- | └─ 12_Historical_vs_Forecast.png
- |
- | └─ Predictive_Analytics_Sales_Forecasting.ipynb
- | └─ README.md
- | └─ Project_Report.pdf

Results

- Successfully trained a Linear Regression model on historical sales data.
 - Generated accurate sales predictions.
 - Forecasted future sales for upcoming months.
 - Identified key factors influencing sales.
 - Visualized business trends using charts and graphs.
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Future Enhancements

- ARIMA Time Series Forecasting
 - SARIMA Forecasting
 - Random Forest Regression
 - XGBoost Regression
 - LSTM Deep Learning Models
 - Real-Time Dashboard Integration
 - Automated Forecasting Pipelines
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Conclusion

This project demonstrates how Predictive Analytics can be applied to historical sales data to forecast future business trends. By leveraging Machine Learning techniques, organizations can improve planning, optimize resources, and make informed decisions based on data-driven insights.

Contact

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